



InsurAI

AI-Powered Full-Stack Insurance Management Platform

3

User Roles

32

REST APIs

13

DB Tables

4,241

Lines React

TEAM-B

Project Overview & Objective

What is InsurAI?

InsurAI is a production-grade, full-stack insurance management platform built during the Infosys Springboard Virtual Internship 6.0.

It serves three distinct roles — Customer, Advisor, and Admin — each with a tailored dashboard and feature set.

The system handles the complete insurance lifecycle: from policy purchase and KYC verification, through claims processing and advisor meetings, to renewal management and AI-powered recommendations.

Built on enterprise-standard technology: Spring Boot 3.2, JWT security, React/Vite frontend, and MySQL 8 with Hibernate JPA.



Customer Portal

Buy, compare, renew, claim, KYC, proposals — self-service from one screen



Advisor Workspace

Meetings, client notes, commission tracking, performance dashboards



Admin Control

Full policy & claims lifecycle, KYC review, ticket management, user CRUD



AI Integration

Groq LLM for document analysis, smart recommendations, and policy chat

Technical Architecture



React Frontend

App.jsx · 4,241 lines · 32 tabs · 3 role dashboards



Spring Boot API

13 controllers · JWT filter · CORS · @PreAuthorize



Service Layer

14 services · underwriting engine · broadcast logic



JPA / Hibernate

12 entities · lazy fetch · @Modifying · cascades



MySQL 8

13 tables · schema + migration_v2 + migration_v3

Security Layer

- JWT (24hr) · BCrypt passwords
- @Lazy circular-dep fix
- @PreAuthorize role checks
- CORS: ports 3000 & 5173
- Stateless SessionCreationPolicy

Key Features



Policy Management

Buy, import, compare 6 plan types, renew with expiry alerts, full detail view with coverage breakdown



Claims & Payments

File claims with 5-stage tracking (Registered→Settlement), drag-drop document upload, full premium history



Advisor System

Broadcast meeting requests to ALL advisors — first to accept auto-declines the rest. Commission tracker included



AI Analysis

Groq API (Llama-3.1-8b) powers chat, document analysis, and the 10-factor recommendation engine



KYC & Proposals

Drag-drop upload for 6 doc types (Aadhaar, PAN, etc.), simulated underwriting with 7-stage proposal lifecycle



Admin Dashboard

User CRUD + role assignment, KYC verify/reject with notes, support ticket status workflow, all-policies view

Three Roles — Tailored Experiences



Customer

14 tabs

- Dashboard + Health Score
- My Policies (detail view)
- Buy / Compare Plans
- File Claims + doc upload
- Pay Premium
- Find & Request Advisor
- KYC Document Upload
- Proposals + Underwriting
- Renewal Alerts
- AI Recommendations
- Support Tickets
- AI Analysis Chat
- Profile
- Renewal History



Advisor

10 tabs

- Dashboard + stats
- Meeting Requests inbox
- Confirmed Meetings
- Performance Tracker
- My Customers
- Customer Policies
- Customer Claims
- Client Notes (tagged)
- Commission Tracker
- AI Analysis



Admin

8 tabs

- Dashboard + stats
- All Users (CRUD)
- All Policies (approve)
- All Claims (review)
- KYC Document Review
- Support Ticket Mgmt
- Reports
- Role Management

8-Week Timeline & Milestones

Week 1-2

Foundation

Developed the frontend modules(login and registration page checking password strength, dashboard for customer ,admin and advisor roles)

Week 3-4

Core Modules

Customer frontend with policies, buy, AI analysis, payments, profile tabs; Admin approves policies/claims; Spring Boot backend updates policies/claims; database stores users, policies, claims, commissions, payments, dashboard shows real-tim data.

Week 5-6

Expansion

Added AI assistance to analyze risks and policies, imported customized policies, completed advisor dashboard, and added feature for customers to request advisor support for policy or claim issues.

Week 7-8

Expansion

Fixed API key, corrected frontend overlaps, improved dashboard, added AI chatbot for queries, enhanced UI with icons/PNGs, added KYC/nominee features, and document submission tab for policy and claim analysis.

KEY MILESTONES



Kickoff

Jan 13



Prototype

Jan 27



Mid-Term

Feb 10



Final Submit

Mar 3



Presentation

Mar 10

Challenges, Learnings & Conclusion

Challenges Faced & Resolved

1

Circular Dependency

AuthService ↔ SecurityConfig — fixed with @Lazy annotation

2

403 on All Endpoints

@Query JOIN FETCH broke Spring Security — removed, used derived query

3

Password Null in JWT

@JsonIgnore on field not getter — moved to getPassword() method

4

Multi-class Java Files

Multiple classes in one file — split each into its own dedicated file

5

JSX Complexity

4,000+ lines JSX — automated brace-balance checks (Node.js script)

Learnings & Skills Acquired

Technical:

Spring Boot (JPA, Security, JWT), React hooks, MySQL migrations, REST design

Domain:

Insurance lifecycle, KYC compliance, underwriting rules, advisor workflows

Practices:

Layered architecture, additive development, DB versioning, defensive coding

Soft Skills:

5-member team collaboration, code review, live demo presentation

Conclusion

InsurAI delivered a fully functional end-to-end insurtech platform in 8 weeks, mirroring production fintech architecture. Every design decision — JWT statelessness, additive v3 migration, broadcast meeting model — reflects industry best practices. The project accelerates each team member's readiness for full-stack or BFSI technology roles.